

ABDUL ALIM RAKIB

AI/ML Engineer | NLP & LLMs | Computer Vision | Full-Stack Web Development

Dhaka, Bangladesh | +880 1748-585134 | abdulalimrakib04@gmail.com
[linkedin.com/in/abdul-alim-rakib1](https://www.linkedin.com/in/abdul-alim-rakib1) | github.com/abdulalimrakib



SUMMARY

AI/ML Engineer with hands-on, end-to-end experience across Natural Language Processing, Computer Vision, and full-stack web development. Fine-tuned Meta-Llama-3.1-8B with QLoRA/PEFT for automated essay evaluation, built a Retrieval-Augmented Generation chatbot over a legal corpus, and shipped a bilingual NLP-driven Android security system. In computer vision, benchmarked 22+ deep learning architectures for image classification and forensic manipulation detection, and deployed a compressed, quantized model to production. Full-stack MERN background enables independent delivery of ML-powered products from model to client, and comfort operating US business hours as part of an international team.

SKILLS

Languages: Python, JavaScript, SQL, C, C++, Java

NLP & LLMs: LoRA/QLoRA fine-tuning, SFT (SFTTrainer), RAG architecture, LangChain, LangGraph, prompt engineering (few-shot, chain-of-thought), Hugging Face Transformers, PEFT, TRL, Unsloth, bilingual (English + Bangla) NLP

Computer Vision: CNNs, Vision Transformers, GANs, image classification, forensic manipulation detection/localization, Grad-CAM, LIME, OpenCV, Albumentations, timm

ML / DL Frameworks: PyTorch, TensorFlow / Keras, Scikit-learn, ONNX / ONNX Runtime, NumPy, Pandas, SciPy

Applied ML Techniques: Knowledge distillation, INT8 quantization, ensemble learning, transfer learning, hyperparameter tuning, explainable AI

Generative AI APIs: OpenAI API, Anthropic Claude API, Gemini API

Web & Backend: React.js, Redux, Node.js, Express.js, MongoDB, REST APIs, JWT authentication, Tailwind CSS

Databases & Tools: MySQL, MongoDB, Vector DB (Chroma), Git / GitHub, Linux CLI, Jupyter Notebook, Postman

PROJECTS

LawBot — RAG Chatbot over Bangladesh Penal Code Corpus

NLP / LLM

- Built an end-to-end Retrieval-Augmented Generation system for natural-language Q&A over a structured legal corpus, one of the first Bangla-domain RAG applications built at the university.
- Implemented the full pipeline: document chunking, embedding generation, vector-store indexing (Chroma), LLM response generation with context injection, and citation-aware output.
- Stack: Python, LangChain, Chroma, OpenAI API, Jupyter Notebook.

IELTS Writing Evaluator — LLM Fine-Tuning (QLoRA/PEFT)

NLP / LLM

- Fine-tuned Meta-Llama-3.1-8B-Instruct using 4-bit QLoRA + LoRA ($r=128$, $\alpha=256$) across all 7 attention/MLP projection layers, trained with TRL's SFTTrainer for 1,500 steps on a curated IELTS Writing Task 2 dataset.
- System scores essays across four IELTS criteria on a 0.5-step band scale with structured, per-criterion corrective feedback.
- Benchmarked against GPT-4.0, Gemini, and DeepSeek on 100 held-out samples using QWK, MAE, Exact Match, and Acc(± 0.5); the fine-tuned model reached competitive agreement with frontier commercial models.
- Optimized training with Unsloth's gradient checkpointing, bfloat16 mixed precision, and a cosine LR schedule with warm-up; saved the LoRA adapter for efficient inference.
- Stack: Hugging Face Transformers, PEFT, TRL, Unsloth, PyTorch, Scikit-learn.

SecureSpeak — Bilingual MFS Threat Detection

NLP / Security

- Designed a context-aware, multi-layer ML security system for Bangladeshi mobile financial users (bKash, Nagad, Rocket), combining supervised classification with real-time threat scoring.
- Built a bilingual (English + Bangla) threat-communication pipeline connecting a Python/REST API ML backend to a Kotlin Android client — full-stack deployment from model to mobile.
- Functioning Android prototype with full API integration complete; preparing a first-author manuscript for academic publication.
- Stack: Python, Scikit-learn, REST API, Kotlin (Android), Jupyter Notebook.

Rice Variety Classification — Benchmark & Edge Deployment

Computer Vision

- Unified three heterogeneous datasets into a 63-class, 98,730-image benchmark; diagnosed and corrected a data-leakage flaw that had inflated reported accuracy by up to 4.8 points.
- Benchmarked 22 pretrained CNN and Vision Transformer architectures under one protocol; best single model (Xception) reached 99.49% accuracy / 98.23% macro-F1, and a 4-model weighted ensemble reached 99.66% / 98.76% via a 420-configuration search.
- Distilled the ensemble into a lightweight ShuffleNetV2 student and replicated a published ensemble-distillation methodology (Hasan et al., IEEE Access 2025) on a new domain, reporting a controlled negative result inconsistent with the source paper's claims.
- Deployed via static per-channel ONNX INT8 quantization (1.69 MB, ~348 images/sec, 96.6% macro-F1); diagnosed and resolved a dynamic-quantization failure (accuracy collapse from 99% to 27%); validated with 5-fold cross-validation, robustness testing, and Grad-CAM / LIME interpretability analysis.
- Stack: PyTorch, timm, Scikit-learn, ONNX / ONNX Runtime, Albumentations.

Deep Forensic Benchmark — Manipulation Detection

Computer Vision

- Implemented and benchmarked state-of-the-art forensic architectures (MVSS-Net++, TruFor, SegFormer B0/B2, EfficientNet-B4, MMFusion) for pixel-level image manipulation detection and localization.
- Prototyped a custom dual-branch ForensicSegNet architecture and constructed manipulation datasets for controlled, reproducible comparative benchmarking.
- Stack: Python, PyTorch, Jupyter Notebook.

Full-Stack Auth Platform — MERN + JWT

Web Development

- Built a production-style authentication platform with role-based access control, JWT session management, bcrypt credential hashing, protected routing, and Redux client-state management.
- Stack: MongoDB, Express.js, React.js, Node.js, Redux, Tailwind CSS.

EXPERIENCE

Undergraduate Teaching Assistant — North South University

Summer 2024 – Summer 2025

- Supported six faculty members across three semesters; graded assignments and enforced consistent rubric application for programming and AI/ML coursework.

Peer Mentor & Moderator — NSU ACM Web Dev Group

Spring 2024 – Spring 2025

- Led a 12-member team and delivered peer-learning workshops on React.js, front-end frameworks, and Git-based workflows.

EDUCATION

B.Sc. in CSE — North South University

Jan 2022 – 2025

- CGPA: 3.65 / 4.00 · Merit-based 25% Tuition Scholarship awarded throughout undergraduate study.
- Core AI/ML coursework: Natural Language Processing, Computer Vision, Neural Networks & Deep Learning, AI Fundamentals.
- CS foundations: Data Structures & Algorithms, DBMS, Software Engineering, Web Application Development.

Higher Secondary Certificate — Notre Dame College

- GPA: 5.00 / 5.00 (national maximum grade).

HONORS & ACTIVITIES

- Best Member Award, NSU ACM Student Chapter — for outstanding technical contribution and peer mentorship.
- Merit-Based Tuition Scholarship (25%), North South University.